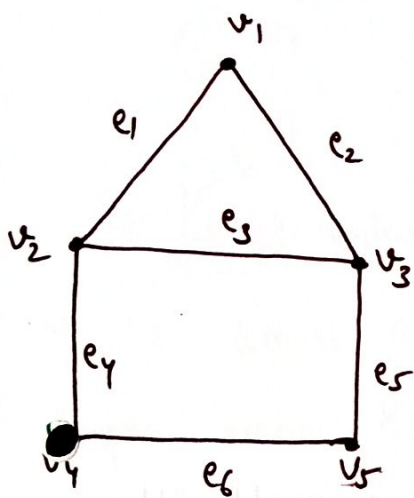


Operations on Graphs

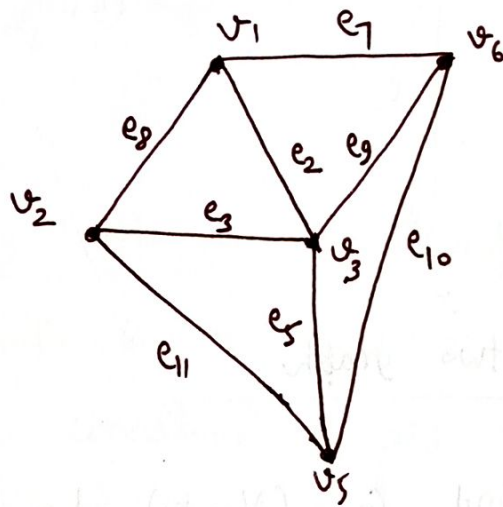
(B)

RTV
 (i) Union of Simple Graph → The union of two simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is a graph $G = (V, E)$, where $V = V_1 \cup V_2$ and $E = E_1 \cup E_2$ and is denoted by ~~$G_1 \cup G_2$~~ $G_1 \cup G_2$.

For ex → Find $G_1 \cup G_2$ where $G_1 = (V_1, E_1) = (\{v_1, v_2, v_3, v_4, v_5\}, \{e_1, e_2, e_3, e_4, e_5, e_6\})$ and $G_2 = (V_2, E_2) = (\{v_1, v_2, v_3, v_4, v_5, v_6\}, \{e_7, e_8, e_9, e_{10}, e_{11}\})$.



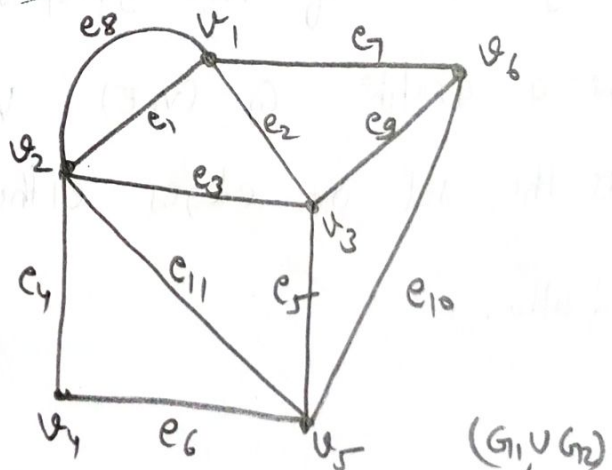
(G₁)



(G₂)

Now $G_1 \cup G_2$ is

$G_1 \cup G_2 = (\{v_1, v_2, v_3, v_4, v_5, v_6\}, \{e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8, e_9, e_{10}, e_{11}\})$.



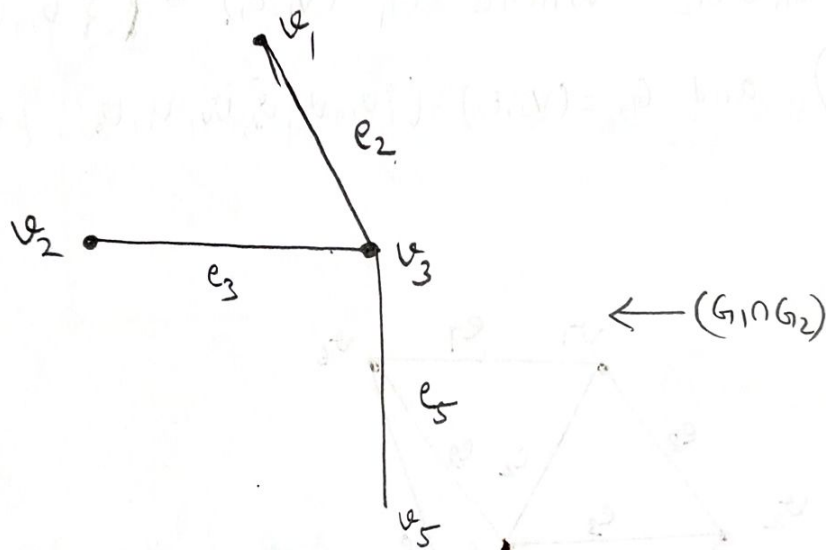
(G₁ ∪ G₂)

(ii) Intersection of Graphs

If $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs then $G_1 \cap G_2$ is denoted by the intersection of two graphs, where $V_1 \cap V_2 = V$ and $E_1 \cap E_2 = E$.

The intersection of above graph is

$$G_1 \cap G_2 = (\{v_1, v_2, v_3, v_5\}, \{e_1, e_3, e_5\})$$

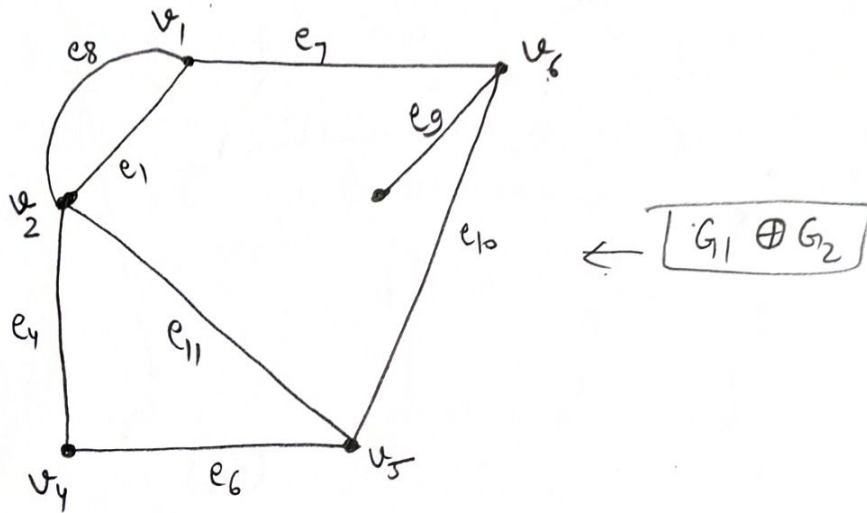


(iii) Ring Sum of two graphs

Let $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ be two graphs then ring sum of two graphs is denoted by $G_1 \oplus G_2$, is a graph $G = (V, E)$ where $V = V_1 \cup V_2$ and E is the set of edges either G_1 or G_2 but not both.

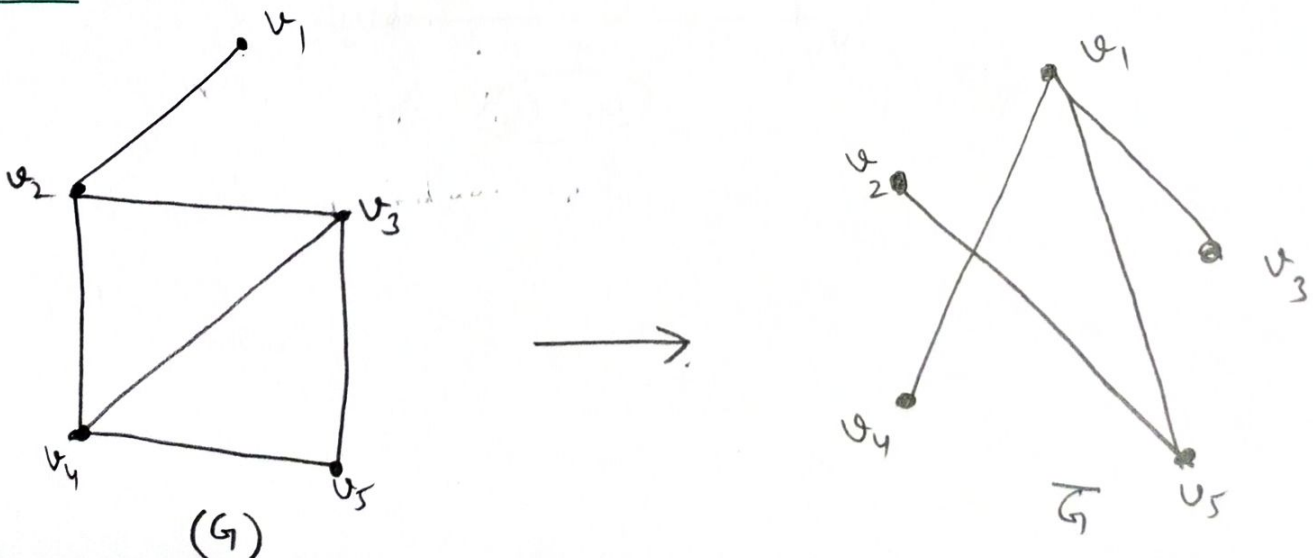
Ex → The ring sum of two graphs G_1 and G_2 as in previous example is $G_1 \oplus G_2$ where $V = V_1 \cup V_2 = \{v_1, v_2, v_3, v_4, v_5, v_6\}$ and $E = \{e_1, e_4, e_5, e_7, e_8, e_9, e_{10}, e_{11}\}$.

The diagram is as follows;



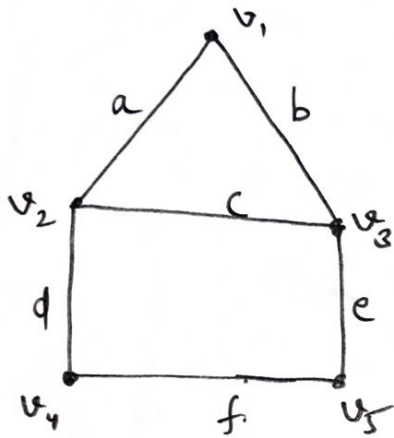
RTU
Complementary Graph → The complementary graph $\bar{G}$ of a simple Graph G on n -vertices, is the subgraph of the complete graph K_n consisting of all n -vertices of G and all edges that are not in G , but are in K_n .

For ex →

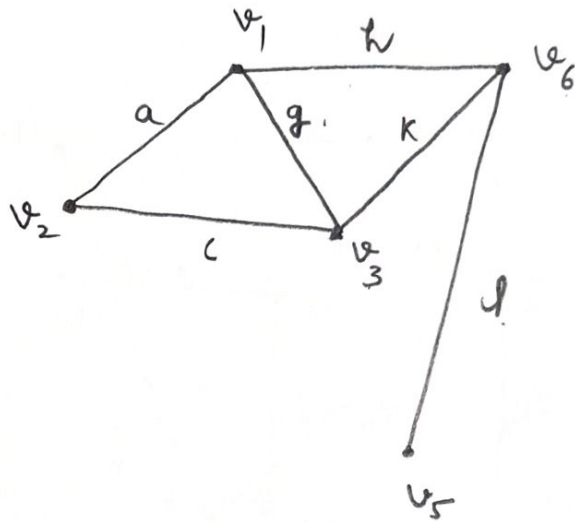


Difference of two graphs $\rightarrow$ The difference of two graphs G_1 and G_2 written as $G_1 - G_2$ is a graph having all those edges which is in G_1 but not in G_2 .

For ex $\rightarrow$

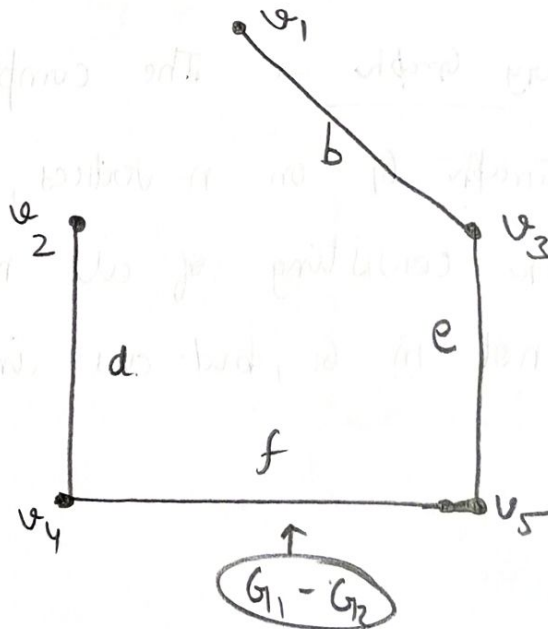


(G_1)



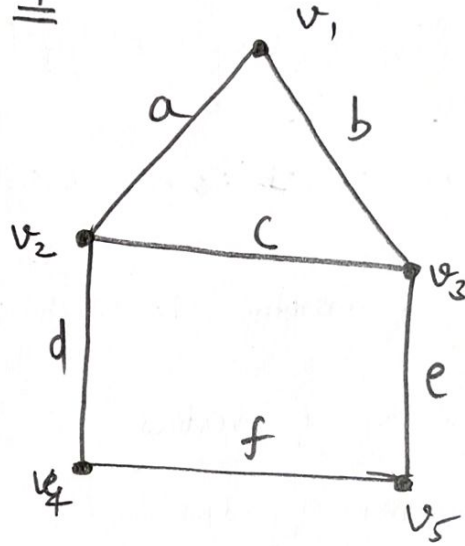
(G_2)

then $G_1 - G_2$ is

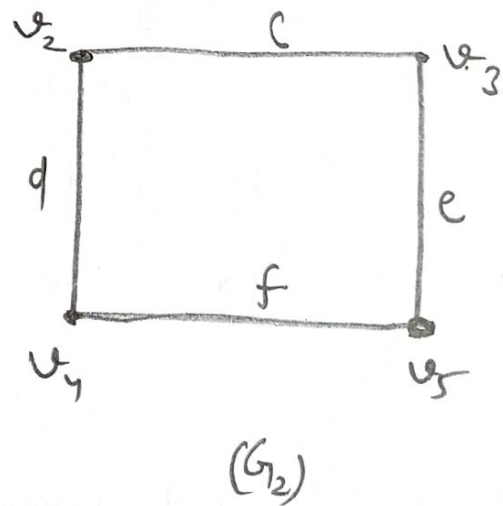
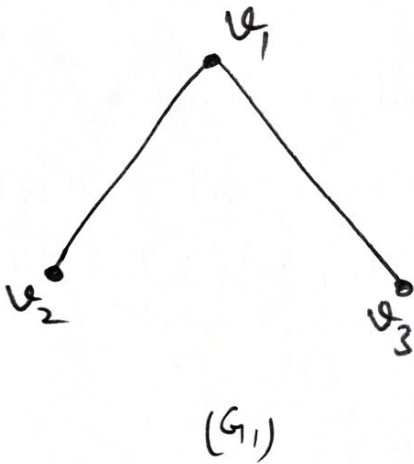


Decomposition of a Graph $\rightarrow$ A graph G is said to be decomposed into two subgraphs G_1 and G_2 if $G_1 \cup G_2 = G$ and $G_1 \cap G_2 = \phi = \text{a null graph}$ (D)

Ex $\rightarrow$ The graph G



Can be decomposed as



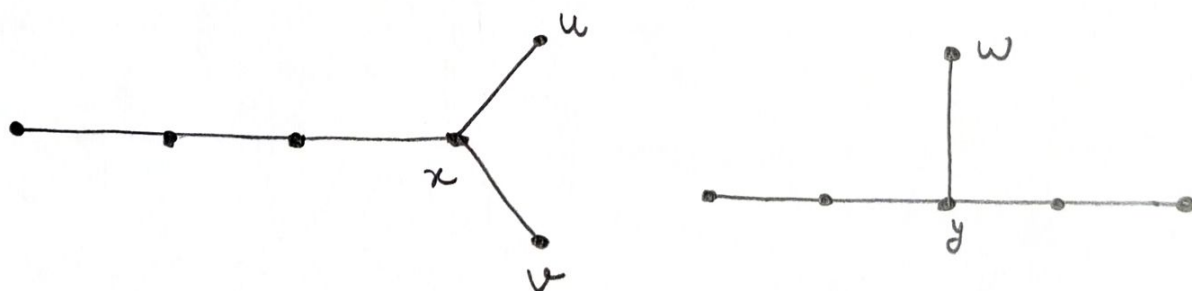
Isomorphic Graph $\rightarrow$ Two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are said to be isomorphic to each other if there exists a bijection mapping f from V_1 to V_2 .

i.e.
 $f: V_1 \rightarrow V_2$ such that for each of the vertices
 $u_i, u_j \in V_1 \Rightarrow \{u_i, u_j\} \in E_1 \Rightarrow \{f(u_i), f(u_j)\} \in E_2$.

The function 'f' is called an isomorphism from G_1 to G_2 .

Two graphs are isomorphic to each other if they must have

- (i) The same number of vertices
- (ii) The same number of edges
- (iii) An equal number of vertices with a given degree i.e. same degree sequence.



The two graphs are not isomorphic to each other because the degree sequence of two graphs are not same.